

DATABASE DESIGN DOCUMENT

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Project Title

PlayStation Slot Booking System

WEEK 2: ENTITY, ATTRIBUTES AND ENTITY RELATIONSHIP DIAGRAM

Entity Identification

1. User

Primary Key (PK): User_ID

Attributes:

- User_ID
- User_Name
- Age
- Gender
- Phone_Number
- Email
- Address
- Password

2. PlayStation Console

Primary Key (PK): Console_ID

Attributes:

- Console_ID
- Console_Name
- Console_Type (PS4 / PS5)
- Hourly_Rate
- Console_Status
- Console_Location

3. Time Slot

Primary Key (PK): Slot_ID

Attributes:

- Slot_ID
- Slot_Date
- Start_Time
- End_Time
- Slot_Status

4. Booking

Primary Key (PK): Booking_ID

Foreign Keys (FK):

- User_ID
- Console_ID
- Slot_ID

Attributes:

- Booking_ID
- Booking_Date
- Session_Duration
- Total_Amount
- Booking_Status

5. Payment

Primary Key (PK): Payment_ID

Foreign Key (FK):

- Booking_ID

Attributes:

- Payment_ID
- Payment_Method
- Payment_Date
- Amount
- Payment_Status
- Transaction_ID

6. Receipt

Primary Key (PK): Receipt_ID

Foreign Key (FK):

- Booking_ID

Attributes:

- Receipt_ID
- Receipt_Date
- Receipt_Number
- Total_Amount

7. Gaming Session

Primary Key (PK): Session_ID

Foreign Keys (FK):

- Booking_ID
- Console_ID

Attributes:

- Session_ID
- Session_Start_Time
- Session_End_Time
- Session_Duration
- Session_Status

8. Admin

Primary Key (PK): Admin_ID

Attributes:

- Admin_ID
- Admin_Name
- Username
- Password
- Email

Relationships

Relationship

User Booking

PlayStation Console Booking

Time Slot Booking

Booking Payment

Booking Receipt

Booking Gaming Session

PlayStation Console Gaming Session

Admin PlayStation Console

Admin Booking

Cardinality

One User can make Many Bookings (1:M)

One Console can have Many Bookings (1:M)

One Time Slot can be reserved for Many Bookings (1:M)

One Booking has One Payment (1:1)

One Booking generates One Receipt (1:1)

One Booking creates One Gaming Session (1:1)

One Console can be used for Many Gaming Sessions (1:M)

One Admin manages Many PlayStation Consoles (1:M)

One Admin manages Many Bookings (1:M)

Entity-Relationship Model

Entities

- User
- PlayStation Console
- Time Slot
- Booking
- Payment
- Receipt
- Gaming Session
- Admin

Primary Keys

- User_ID
- Console_ID
- Slot_ID
- Booking_ID
- Payment_ID
- Receipt_ID

- Session_ID
- Admin_ID

Foreign Keys

- Booking User_ID
- Booking Console_ID
- Booking Slot_ID
- Payment Booking_ID
- Receipt Booking_ID
- Gaming Session Booking_ID
- Gaming Session Console_ID

Database Description

The PlayStation Slot Booking System is designed to automate the process of reserving PlayStation gaming sessions in a gaming lounge. Customers can register, log in, view available PlayStation consoles, select an available date and time slot, choose the duration of the session, complete the payment process, and receive a booking receipt. The administrator is responsible for managing PlayStation consoles, bookings, and customer records. The database stores all booking, payment, session, and receipt information securely while maintaining relationships between entities using primary and foreign keys. This design minimizes data redundancy, improves data integrity, and provides an efficient system for managing gaming slot reservations.

Entity-Relationship Model:

ER DIAGRAM – PLAYSTATION SLOT BOOKING SYSTEM

